

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Construction and working of the Reciprocating pump with neat sketch.

Q.24 Explain Plate and frame filter press with neat sketch.

Q.25 Explain Carnot cycle and how its efficiency is calculated.

No. of Printed Pages : 4

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3rd Sem / Plastic Technology

Subject : Basics of Chemical Engineering / Fluid Flow / Viscose Flow of Fluids / Unit Operations - I

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is a type of coarse crusher?

- a) Roll mill b) Stamp battery
- c) Jaw crusher d) Dicer

Q.2 For an ideal black body_____

- a) Absorptivity = 1 b) Reflectivity = 1
- c) Emissivity = 0 d) Transmissivity = 1

Q.3 Heat transfer takes place according to which law?

- a) Newton's law of cooling
- b) Second law of thermodynamics
- c) Newton's second law of motion
- d) First law of thermodynamics

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Q.4 The transfer of heat between two bodies, which are in direct contact is called

- a) Radiation
- b) convection
- c) conduction
- d) none of the mentioned above

Q.5 The efficiency of the ideal screens is _____

- a) Unity b) Infinity
- c) zero d) Can't be determined

Q.6 Which of the following works principle of compression?

- a) Rod mill b) Knife cutter
- c) Blake jaw crusher d) Gyratory Crusher

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define Heat transfer operation.

Q.8 Define mesh number.

Q.9 Define Grinding.

Q.10 Define Fick's law.

Q.11 Define conduction.

Q.12 The value of C_p is always less than C_v (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Describe the First law of thermodynamics with example.

Q.14 Draw the sketch of cyclone separator explains its working.

Q.15 Explain open and closed system with example.

Q.16 Define conduction and convection with suitable examples.

Q.17 Explain the free and Forced convection with example.

Q.18 Explain Fourier law of heat conduction for single conductive wall.

Q.19 Explain Isochoric process of thermodynamics.

Q.20 Write the difference between crushing and grinding.

Q.21 Explain cyclones Separators with diagram.

Q.22 Explain in detail solenoid valve.

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